

Pearson Edexcel International GCSE

Mathematics A

Modular Unit 2

November 2024 Reconstructed Past Paper

STUDENT WORKBOOK

Adaptive working-space edition

35 adaptive question sections 121 marks

Selected from Linear Higher Papers 1H and 2H

Dr Eslam Ahmed

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L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

Assistant Lecturer, Faculty of Engineering, Cairo University

B.Sc. Electrical Engineering (2017), Distinction with Honours

M.Sc. Mathematics (2023)

PhD Candidate

eliteigcse.com • WhatsApp +20 112 000 9622

About this reconstructed paper

Each question section keeps the exact source demand beside practical working space. A source grid, graph, table, or diagram is retained once at a usable size.

Ownership rule: Unit 2 owns a demand when its assessed or terminal statement is Unit 2, even if Unit 1 skills are prerequisites. For example, drawing boundary lines supports the Unit 2 region-of-inequalities demand. This book contains the demands owned by Unit 2; the selection contains 121 marks.

This is an Elite reconstructed teaching paper selected from official Linear Higher material; it is not an official Pearson Modular examination paper. Source assets are retained for private teaching use.

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Question 01

4MA1-1H Q1 M01

MODULAR UNIT 2

1 marks

- 1 The table shows some information about the hourly rates of pay of 60 workers.

Hourly rate of pay (p dollars)	Frequency
$10 < p \leq 15$	18
$15 < p \leq 20$	16
$20 < p \leq 25$	14
$25 < p \leq 30$	8
$30 < p \leq 35$	4

- (a) Write down the modal class.

.....
(1)

WORKING SPACE

Question 02

4MA1-1H Q1 M02

MODULAR UNIT 2

4 marks

- 1 The table shows some information about the hourly rates of pay of 60 workers.

Hourly rate of pay (p dollars)	Frequency
$10 < p \leq 15$	18
$15 < p \leq 20$	16
$20 < p \leq 25$	14
$25 < p \leq 30$	8
$30 < p \leq 35$	4

- (b) Work out an estimate for the mean hourly rate of pay of the 60 workers.

..... dollars
(4)

WORKING SPACE

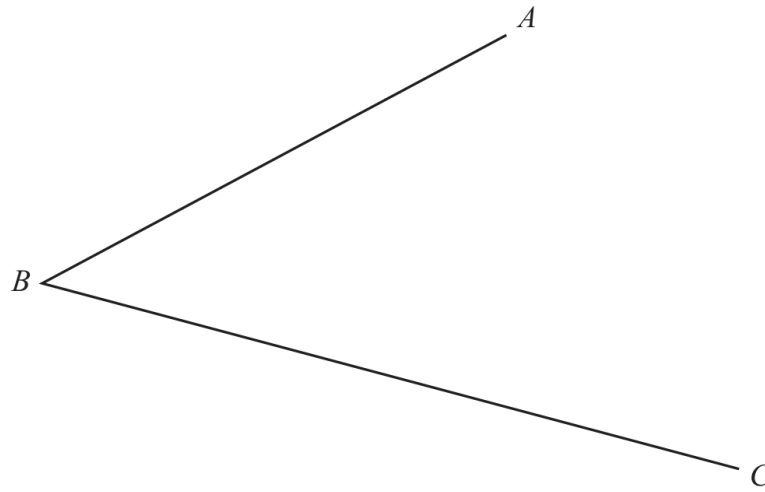
Question 03

4MA1-1H Q2 M01

MODULAR UNIT 2

2 marks

- 2 Use ruler and compasses only to construct the bisector of angle ABC
You must show all your construction lines.



Question 04

4MA1-1H Q5 M01

MODULAR UNIT 2

4 marks

- 5 The diagram shows a cylinder.

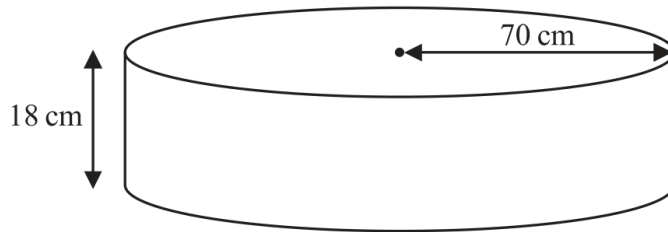


Diagram **NOT**
accurately drawn

The radius of the cylinder is 70 cm

The height of the cylinder is 18 cm

Work out the volume of the cylinder.

Give your answer in litres correct to the nearest litre.

..... litres

WORKING SPACE

Question 05

4MA1-1H Q6 M01

MODULAR UNIT 2

2 marks

6 $A = 2^3 \times 5^4 \times 7 \times 11$

$B = 2^2 \times 5^2 \times 7^2$

$C = 2^2 \times 5^3 \times 7^4$

Find the highest common factor (HCF) of A , B and C
Write your answer as a product of prime factors.

.....

WORKING SPACE

Question 06

4MA1-1H Q7 M01

MODULAR UNIT 2

4 marks

- 7 Shop **A** and Shop **B** have offers for buying the same type of laptop.

The normal price of the laptop in Shop **A** is different to the normal price of the laptop in Shop **B**

Shop A	Shop B
Our normal price £475	Get 15% off our normal price
Get 16% off our normal price	Only pay £408

Which shop gives more money off their normal price?
Show your working clearly.

WORKING SPACE

Question 07

4MA1-1H Q8 M03

MODULAR UNIT 2

3 marks

(b) Solve the inequality $3y + 5 > 7y - 10$

Show clear algebraic working.

.....
(3)

WORKING SPACE

Question 08

4MA1-1H Q9 M01

MODULAR UNIT 2

1 marks

- 9 (a) Write 8.4×10^{-5} as an ordinary number.

.....
(1)

WORKING SPACE

Question 09

4MA1-1H Q9 M02

MODULAR UNIT 2

2 marks

- (b) Work out $(6.5 \times 10^{-40}) \times (8 \times 10^{185})$
Give your answer in standard form.

.....
(2)

WORKING SPACE

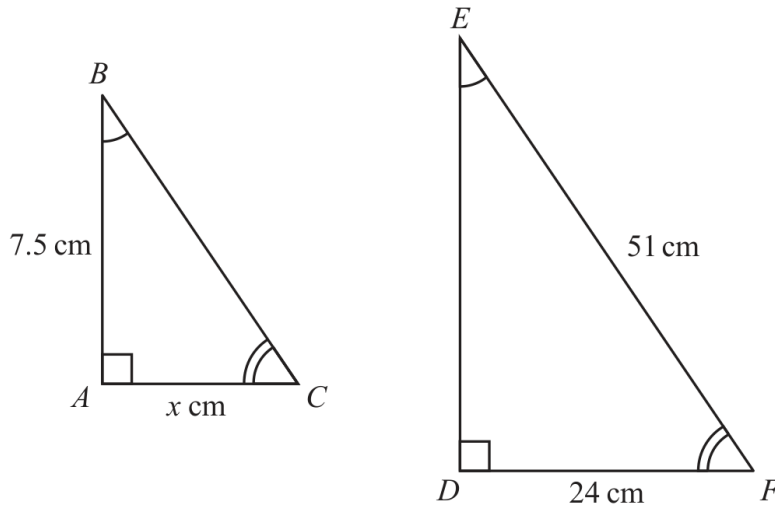
Question 10

4MA1-1H Q10 M01

MODULAR UNIT 2

5 marks

10 Here are two similar triangles.



Diagrams **NOT**
accurately drawn

Work out the value of x
Show your working clearly.

$x =$

WORKING SPACE

Question 11

4MA1-1H Q14 M01

MODULAR UNIT 2

3 marks

14 B is inversely proportional to the square of d

$$B = 0.25 \text{ when } d = 12$$

Find a formula for B in terms of d

.....

WORKING SPACE

Question 12

4MA1-1H Q19 M01

MODULAR UNIT 2

5 marks

19 An arithmetic series has first term a and common difference d

The sum of the first 30 terms of the arithmetic series is 4395

The sum of the 10th term and the 20th term is 284

Work out the sum of the first 45 terms of the arithmetic series.
Show clear algebraic working.

.....

WORKING SPACE

Question 13

4MA1-1H Q20 M01

MODULAR UNIT 2

4 marks

20 The curve with equation $y = 2x^4 - 64x$ has a minimum point.

Find an equation of the tangent to the curve at the minimum point.

Show clear algebraic working.

WORKING SPACE

Question 14

4MA1-1H Q22 M01

MODULAR UNIT 2

3 marks

- 22 The diagram shows a bowl in the shape of a hemisphere.
The bowl is made from metal.

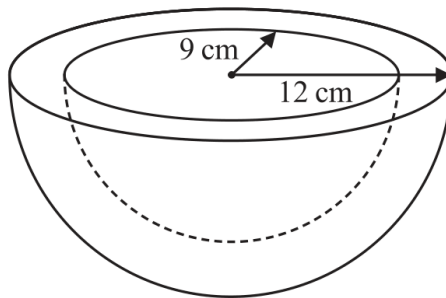


Diagram **NOT**
accurately drawn

The outer radius of the bowl is 12 cm
The inner radius of the bowl is 9 cm

The thickness of the bowl is uniform.

Work out the volume of the metal.
Give your answer correct to the nearest whole number.

..... cm³

WORKING SPACE

Question 15

4MA1-1H Q23 M01

MODULAR UNIT 2

5 marks

- 23 The curve **C** has equation $y = x^2 - 8x - 9$
The straight line **L** has equation $y = k$ where k is an integer.

C and **L** intersect at the points A and B

The coordinates of point A are (p, k)

The coordinates of point B are (q, k)

Given that $p - q = 14$

find the value of k

Show clear algebraic working.

Question 15 (continued)

4MA1-1H Q23 M01

MODULAR UNIT 2

5 marks

$k = \dots\dots\dots$

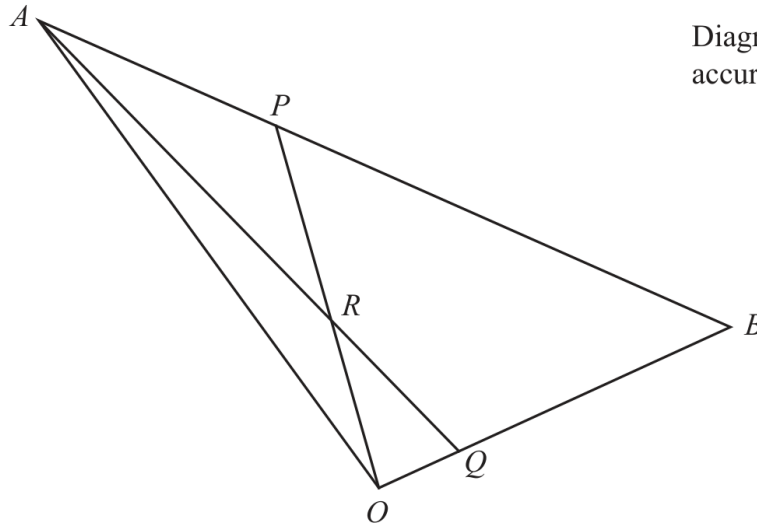
WORKING SPACE

Question 16

4MA1-1H Q24 M01, M02, M03

MODULAR UNIT 2

6 marks

24 OAB is a triangle.Diagram **NOT**
accurately drawn

$$\vec{OA} = 10\mathbf{a} \quad \vec{OB} = 10\mathbf{b}$$

 ARQ and ORP are straight lines.

$$\vec{AP} = \frac{1}{4} \vec{AB} \quad \text{and} \quad \vec{OQ} = \frac{1}{5} \vec{OB}$$

Write the following vectors in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answers.

(i) $\vec{AQ}$

.....
(1)

(ii) $\vec{OP}$

.....
(1)

Question 16 (continued)

4MA1-1H Q24 M01, M02, M03

MODULAR UNIT 2

6 marks

(iii) $\vec{OR}$
(4)

WORKING SPACE

Question 17

4MA1-1H Q25 M01

MODULAR UNIT 2

4 marks

25 The function f is such that $f(x) = 2x^2 - 24x + 7$ where $x \geq 6$

Find the inverse function $f^{-1}(x)$

$f^{-1}(x) = \dots\dots\dots$

WORKING SPACE

Question 18

4MA1-2H Q4 M01

MODULAR UNIT 2

2 marks

- 4 N is a number.
17% of N is 357

(a) Work out the value of N

$N = \dots\dots\dots$
(2)

WORKING SPACE

Question 19

4MA1-2H Q4 M02

MODULAR UNIT 2

3 marks

In 2019, the population of a village was 650

In 2020, the population of the village was 806

(b) Work out the percentage increase in the population.

..... %
(3)

WORKING SPACE

Question 20

4MA1-2H Q6 M01

MODULAR UNIT 2

3 marks

- 6 The diagram shows a solid triangular prism.

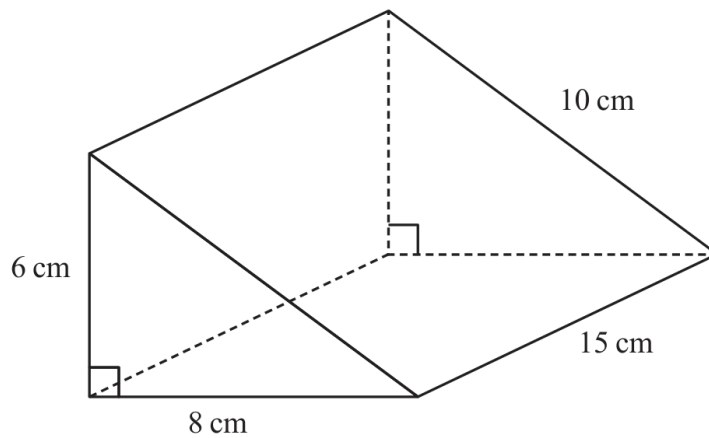


Diagram **NOT**
accurately drawn

Work out the **total** surface area of the triangular prism.

..... cm^2

WORKING SPACE

Question 21

4MA1-2H Q7 M01, M02

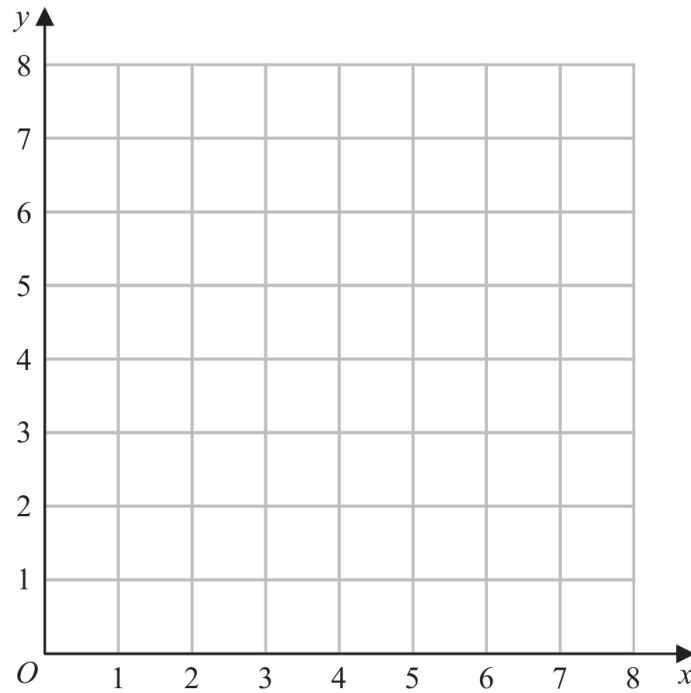
MODULAR UNIT 2

4 marks

7 (a) On the grid, draw the straight line with equation

(i) $x = 3$ (ii) $y = 1$ (iii) $x + y = 7$

Label each line with its equation.



(3)

(b) Show, by shading on the grid, the region that satisfies all three of the inequalities

$$x \geq 3 \quad y \geq 1 \quad x + y \leq 7$$

Label the region **R**

(1)

Question 22

4MA1-2H Q8 M01

MODULAR UNIT 2

3 marks

- 8 Kim puts 4 bananas in a bag.
The mean weight of the 4 bananas in the bag is 145 grams.

Andy puts one more banana into the bag.
The mean weight of the 5 bananas in the bag is 142 grams.

Work out the weight of the banana that Andy puts into the bag.

..... grams

WORKING SPACE

Question 23

4MA1-2H Q9 M01

MODULAR UNIT 2

3 marks

- 9 Nisha invests 20 000 euros for 3 years in a savings account.
She gets 3.5% per year compound interest.

Work out how much money Nisha will have in her savings account
at the end of the 3 years.

Give your answer correct to the nearest euro.

..... euros

WORKING SPACE

Question 24

4MA1-2H Q10 M01

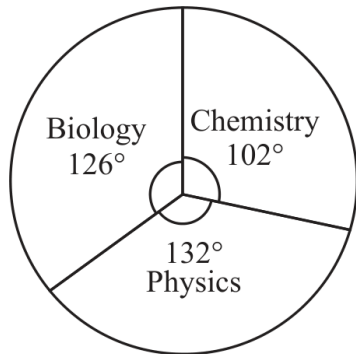
MODULAR UNIT 2

5 marks

- 10 All the students in year 10 and all the students in year 11 named their favourite science subject from Biology, Chemistry and Physics.

The pie chart shows information about the results for the year 10 students.

The table shows information about the results for the year 11 students.



Pie chart for year 10

science subject	number of students
Biology	$3x + 6$
Chemistry	$5x + 8$
Physics	$7x - 9$

Table for year 11

There are 300 students in year 10

There are 320 students in year 11

More students in year 10 than in year 11 said Biology was their favourite science subject.

How many more?

Question 24 (continued)

4MA1-2H Q10 M01

MODULAR UNIT 2

5 marks

WORKING SPACE

Question 25

4MA1-2H Q11 M01

MODULAR UNIT 2

5 marks

11

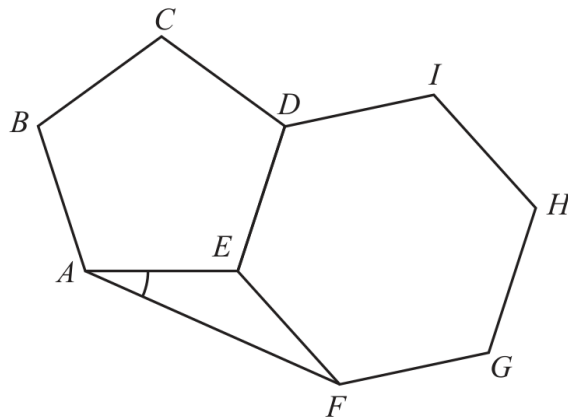


Diagram **NOT**
accurately drawn

$ABCDE$ is a regular pentagon.

$DEFGHI$ is a regular hexagon.

AF is a straight line.

Work out the size of angle EAF

o

Working space for Question 25

Continue your method

UNIT 2

5 marks

WORKING SPACE

Question 26

4MA1-2H Q12 M02

MODULAR UNIT 2

4 marks

(b) Make c the subject of the formula $f = \sqrt{\frac{a+bc}{c-d}}$

.....
(4)

WORKING SPACE

Question 27

4MA1-2H Q13 M01, M02, M03, M04

MODULAR UNIT 2

6 marks

- 13 The frequency table gives information about the weights, in kilograms, of 60 parcels in a delivery van.

Weight (w kilograms)	Frequency
$0 < w \leq 1$	4
$1 < w \leq 2$	15
$2 < w \leq 3$	20
$3 < w \leq 4$	11
$4 < w \leq 5$	6
$5 < w \leq 6$	4

- (a) Complete the cumulative frequency table.

Weight (w kilograms)	Cumulative frequency
$0 < w \leq 1$	
$0 < w \leq 2$	
$0 < w \leq 3$	
$0 < w \leq 4$	
$0 < w \leq 5$	
$0 < w \leq 6$	

(1)

- (b) On the grid opposite, draw a cumulative frequency graph for your table.

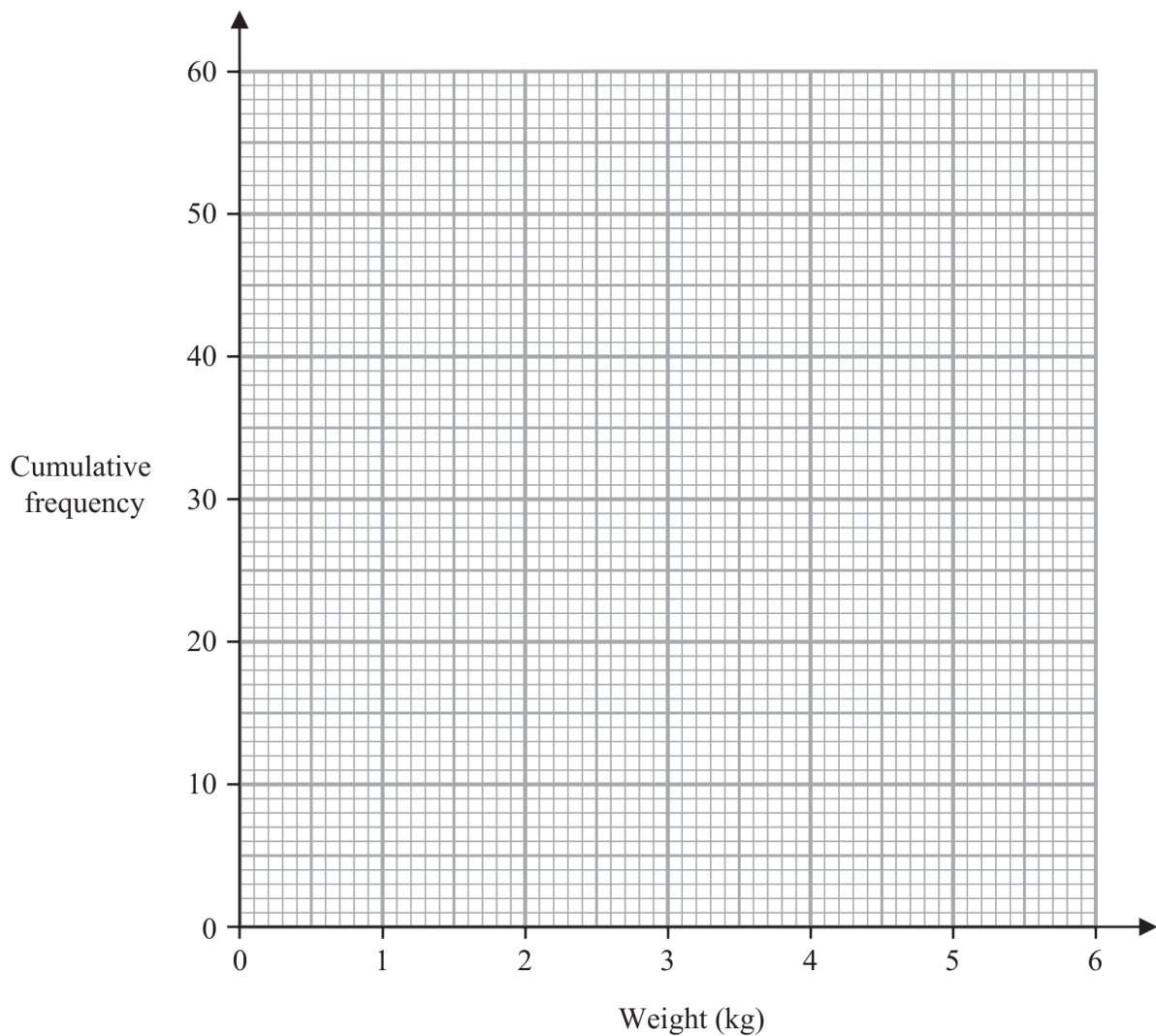
(2)

Question 27 (continued)

4MA1-2H Q13 M01, M02, M03, M04

MODULAR UNIT 2

6 marks



(c) Use your graph to find an estimate for the median weight of the 60 parcels.

..... kilograms
(1)

(d) Use your graph to find an estimate for the number of these parcels that weigh more than 3.7 kilograms.

.....
(2)

Question 28

4MA1-2H Q16 M01, M02, M03, M04

MODULAR UNIT 2

5 marks

16 A, B, C and D are points on a circle, centre O

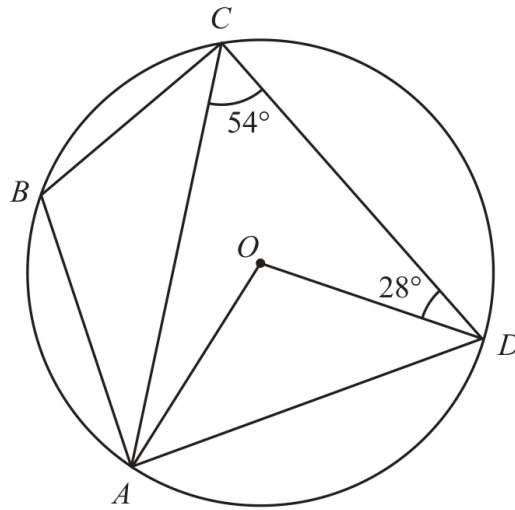


Diagram **NOT**
accurately drawn

(a) (i) Work out the size of angle AOD

.....
(1)

(ii) Give a reason for your answer to part (a)(i)

.....
.....
(1)

(b) Work out the size of angle CAO

.....
(1)

(c) Work out the size of angle ABC

.....
(2)

Working space for Question 28

Continue your method

UNIT 2

5 marks

WORKING SPACE

Question 29

4MA1-2H Q17 M01

MODULAR UNIT 2

3 marks

17 Here are two vectors.

$$\vec{FG} = \begin{pmatrix} -5 \\ 2 \end{pmatrix} \quad \vec{HG} = \begin{pmatrix} 4 \\ 14 \end{pmatrix}$$

Calculate the magnitude of the vector $\vec{HF}$

.....

WORKING SPACE

Question 30

4MA1-2H Q19 M01

MODULAR UNIT 2

1 marks

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.

The coordinates of this minimum point are (5, 4)

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 5)$

(.....,)
(1)

WORKING SPACE

Question 31

4MA1-2H Q19 M02

MODULAR UNIT 2

1 marks

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.
The coordinates of this minimum point are (5, 4)

(ii) $y = 3f(x)$

(..... ,)
(1)

WORKING SPACE

Question 32

4MA1-2H Q19 M03

MODULAR UNIT 2

1 marks

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.

The coordinates of this minimum point are (5, 4)

(iii) $y = f(x) - 7$

(.....,)
(1)

WORKING SPACE

Question 33

4MA1-2H Q20 M01

MODULAR UNIT 2

3 marks

- 20 Solve the inequality $10x^2 + 11x - 21 < 0$
Show clear algebraic working.

WORKING SPACE

Question 34

4MA1-2H Q22 M01

MODULAR UNIT 2

5 marks

22 Solve the simultaneous equations

$$x^2 + y^2 + y = 3$$

$$x + 2 = y$$

Show clear algebraic working.

.....

WORKING SPACE

Question 35

4MA1-2H Q24 M01

MODULAR UNIT 2

6 marks

24 A , B and C are three points on horizontal ground.

$$AB = 8.4 \text{ metres} \quad BC = 9.2 \text{ metres}$$

B is on a bearing of 067° from A

C is on a bearing of 129° from B

Calculate the bearing of A from C

Give your answer correct to the nearest degree.

Question 35 (continued)

4MA1-2H Q24 M01

MODULAR UNIT 2

6 marks

.....

WORKING SPACE